



Project Proposal: BotaniScan

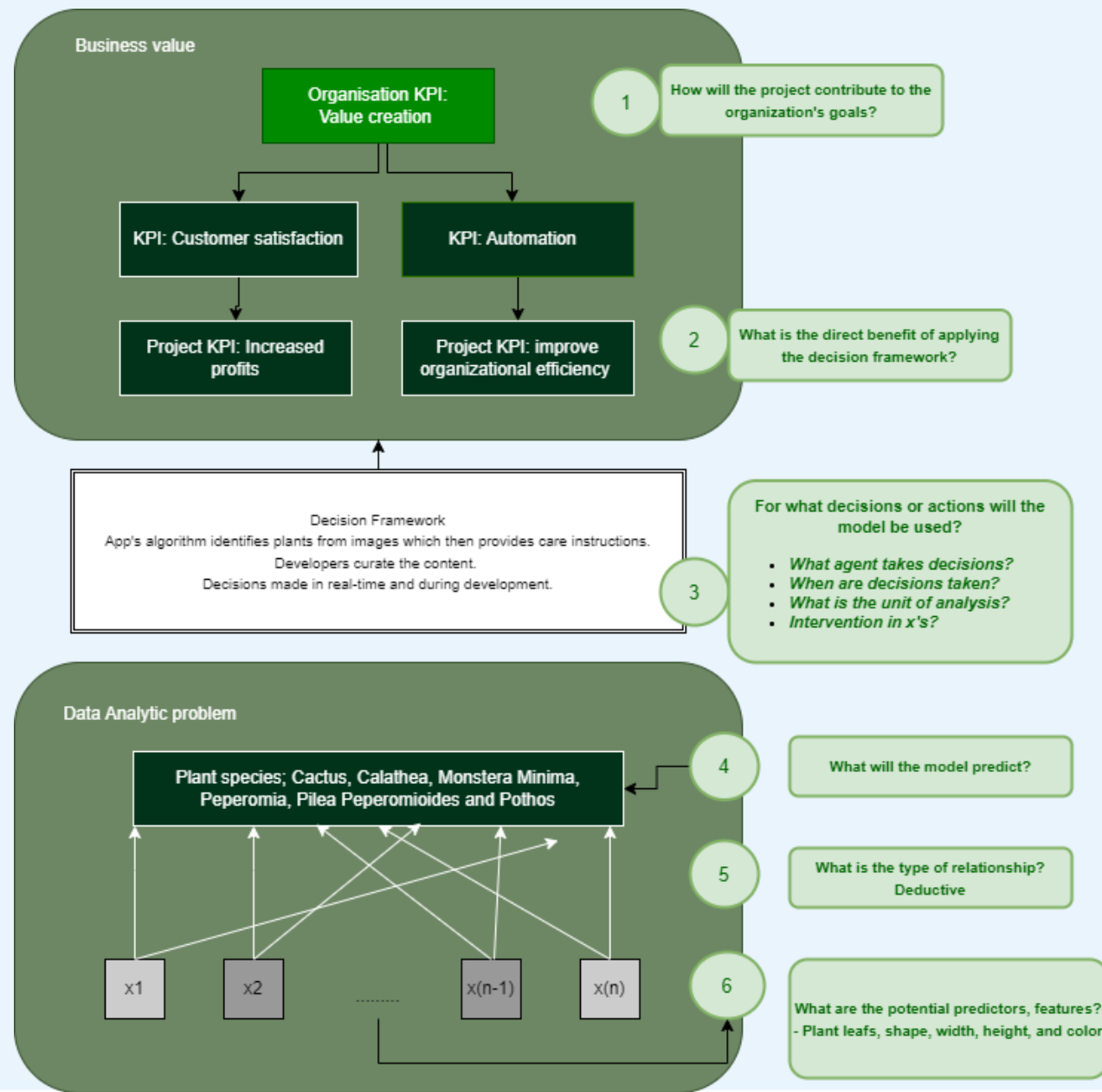
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DISCOVER YOUR WORLD

Project Pitch

Business Understanding

- Create the first idea for your application:
 - Competitive market
 - Botanical research centers, plant shops and customers/consumers
 - A user-friendly customer-faced feature



Problem Overview

Deep Learning

- Explain your image classification problem:
 - Plant identifier
 - 6 classes (Cactus, Calathea, Monstera Minima, Peperomia, Pilea Peperomioides, and Pothos)
 - 1200 images – 200 each
 - Random Guess Accuracy: 17%
 - Human Level Performance Accuracy: 77%
 - Basic MLP: 27%



Pothos



Pilea Peperomioides



Peperomia



Monstera Minima



Calathea



Cactus

Model Overview

Deep Learning

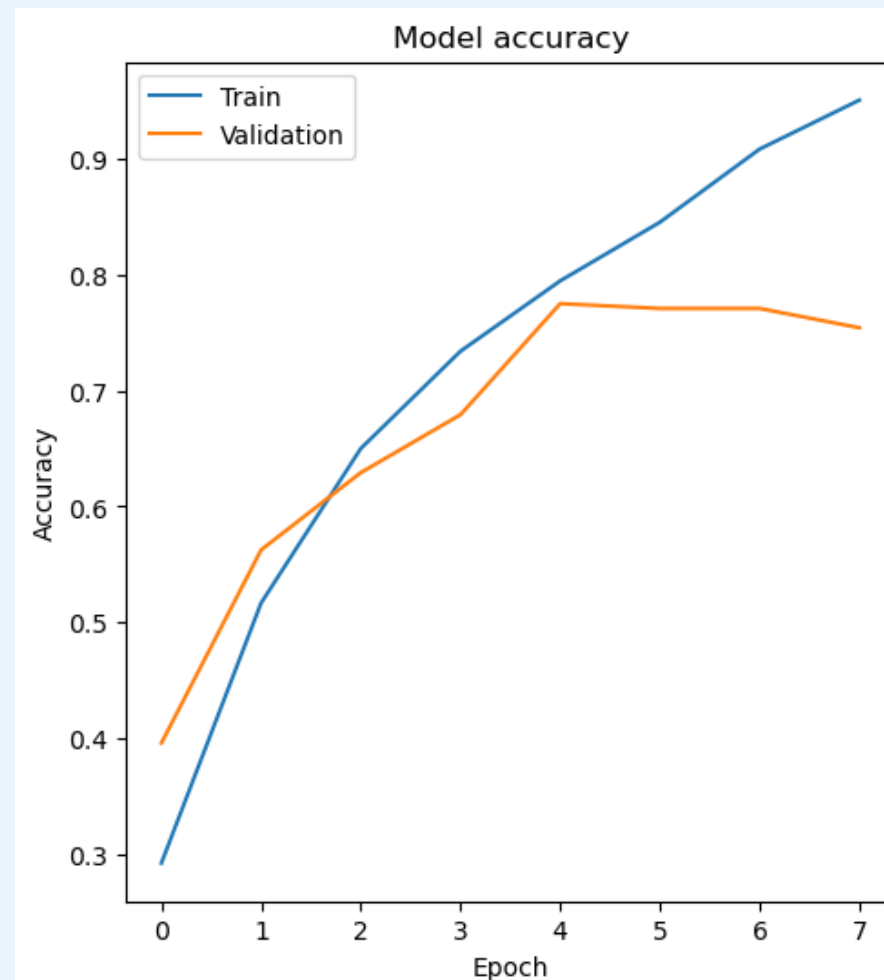
- Explain the process of creating and training your image classification model:
 - Iteration 1: Normal CNN (BatchNorm, Dropout, MaxPooling, and Conv2D)
 - Iteration 2: Data Augmentation
 - Iteration 3: Transfer Learning
 - Iteration 4: Data Augmentation and Transfer Learning

Model Performance

Deep Learning

- Present the performance of your model:
 - Accuracy of 83%

MLP:	Final Iteration:
Confusion Matrix:	Confusion Matrix:
[[6 1 4 0 1 2]	[[14 0 0 0 0 0]
[3 8 3 1 0 7]	[1 19 1 1 0 0]
[4 5 6 0 1 5]	[0 1 17 1 0 2]
[1 7 4 1 0 8]	[1 3 0 14 3 0]
[3 6 4 0 2 9]	[0 0 0 0 21 3]
[1 2 6 0 0 9]]	[0 0 3 1 0 14]]
Accuracy: 0.2667	Accuracy: 0.8250
Precision: 0.3496	Precision: 0.8248
Recall: 0.2667	Recall: 0.8250
F1 Score: 0.2354	F1 Score: 0.8228



Model Interpretability

Responsible AI

- Discuss the trade-off between accuracy and interpretability:
 - Accuracy ensures reliable information
 - Interpretability ensures trust and understanding

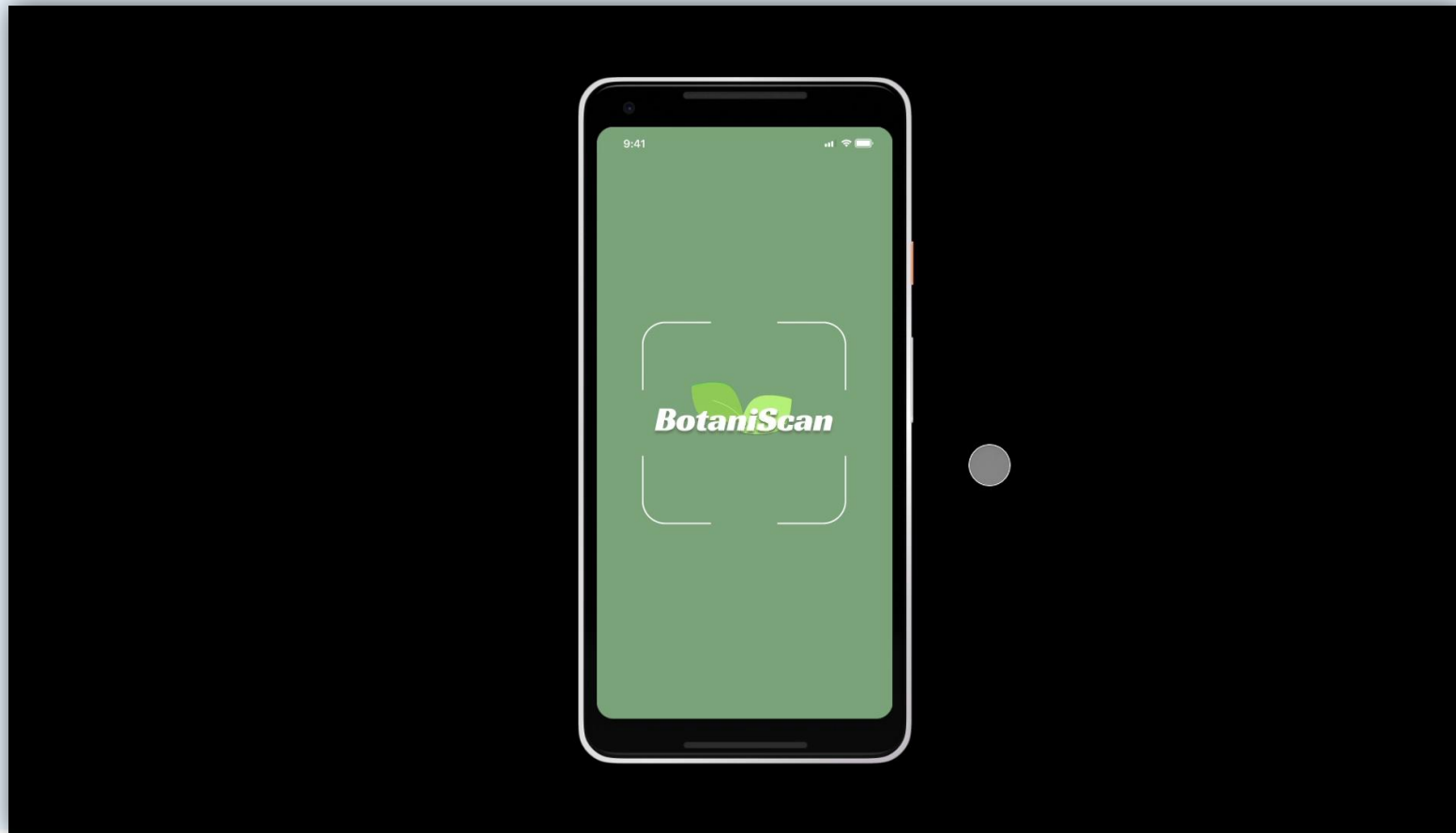
User Study

Human-Centered AI

- A/B Test
- 36 responses
- Removed, added, improved

Demo

Human-Centered AI [link](#)



Thank you!

Any questions?